Atmospheric conditions and fast ice at Davis, East Antarctica: A case study

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[1] We present in situ observations of atmospheric and fast-ice parameters at Davis (68°35′S, 77°58′E), East Antarctica. Surface measurements from 1969–2003 reveal negative trends for summer and autumn surface-air temperatures, yet positive annual, winter and spring trends. Mean sea level pressure decreased over the observational record both annually (-0.10 hPa yr^{-1}, significant at the 90% confidence level (90)) and seasonally. We mainly associate this with increased cyclonic activity. These trends reflect large-scale circulation changes identified by others in the troposphere and lower stratosphere. Temporal changes in the Davis landfast-ice characteristics are evaluated using in situ measurements from the late 1950s together with intermittent data from 1979 onward. There is large interannual variability in fast-ice characteristics, which peaked during the 1990s. Mean values of annual maximum ice thickness taken over subintervals have changed little. However, dates of annual maximum ice thickness and final fast-ice breakout are delayed (each by +0.43 d yr^{-1}). Delays in ice breakout contribute to a prolonged persistence of the fast ice (+0.67 d yr^{-1}, (90)). Linear regression was applied to relate changes in fast-ice characteristics to those in the atmosphere. Significant at the 90% level or above are: Winter temperature and annual maximum ice thickness are anticorrelated ($R^2 = -0.69$); the tendency toward later dates of annual maximum ice thickness appears influenced by winter ($R^2 = +0.47$) and spring ($R^2 = +0.42$) warming. Interannual changes in spectral density of the (annual and summer, respectively) mean sea level pressure within the cyclone-active frequency band are crucial to annual maximum ice thickness ($R^2 = -0.54$) and fast-ice breakout ($R^2 = -0.59$). Maximum ice thickness is furthermore influenced by winter wind speed ($R^2 = -0.52$).

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1. Introduction

[2] Predictions of changes in the polar climate depend on our ability to simulate accurately sea ice and its role within the climate system [*Manabe et al.*, 1991]. Ice thickness is an important parameter to describe the sea-ice state. Sea ice and its thickness evolution are influenced by thermodynamic as well as by dynamic processes [e.g., *Hibler and Ackley*, 1983; *Zhang et al.*, 2000]. For the pack ice, however, it is difficult to directly link variability in ice characteristics with changes in thermodynamics or dynamics due to the numerous interactions between ocean, sea ice and atmosphere.

[3] Temporal variability and trends in the polar atmosphere of the Southern Hemisphere have been identified, for example, changes in latitude and frequency of extratropical cyclone trajectories [*Jones and Simmonds*, 1993; *Simmonds and Keay*, 2000], or changes in the coastal precipitation pattern [*Budd et al.*, 1995; *Smith et al.*, 1998]. Studies have been carried out to determine causal links between atmosphere and pack-ice changes [e.g., *Simmonds*, 1996]. Past research focused on the interactions between atmosphere and pack ice, and considered parameters such as pack-ice extent or concentration. The relocation of the northern sea-ice edge due to changes in cyclone strength and also due to relocation of the polar low-pressure trough has been identified as a dominant atmosphere-ice coupling [*Jones and Simmonds*, 1993]. Changes in sea-ice concentration have been linked to changes in the spatial distribution of the net precipitation [*Watkins and Simmonds*, 2000]. To supplement the research on interactions between pack ice and atmosphere this study investigates the direct interactions between fast ice and atmosphere as derived from in situ observations at Davis on the East Antarctic coast.

[4] Each austral autumn land-fast sea ice forms around much of the Antarctic continent, while pack ice covers the Southern Ocean's seasonal ice zone away from the coastal regions. At its maximum, fast ice can make up as much as 14% by area of the East Antarctic sea ice [*Fedotov et al.*, 1998]. The immobility of the fast ice makes it an ideal tool to study ocean, ice and atmosphere interactions with focus

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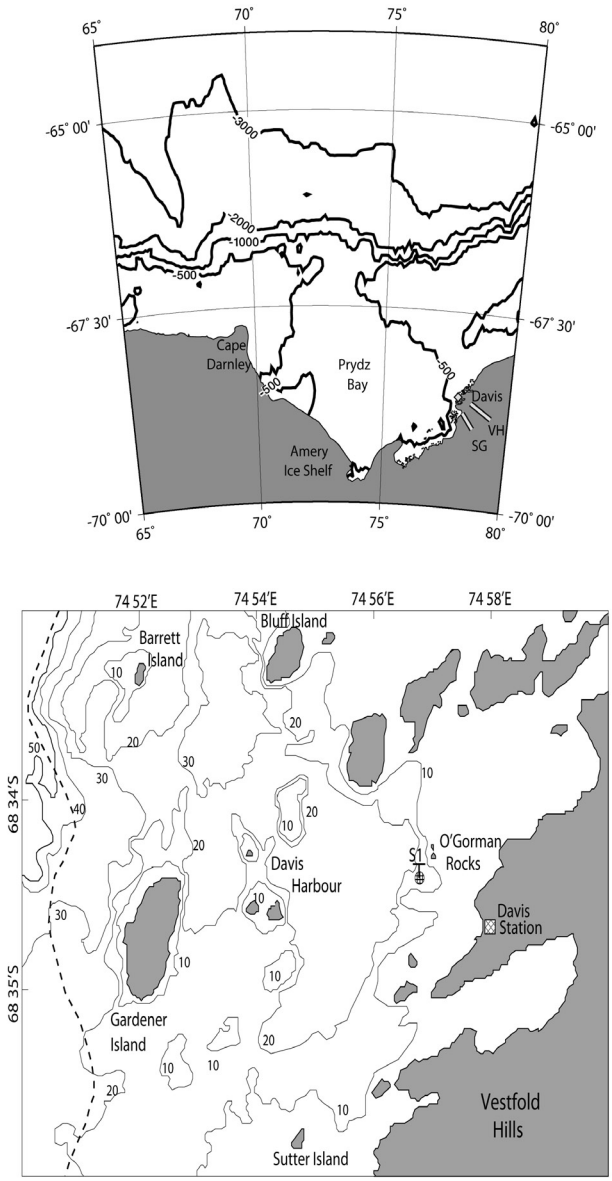

Figure 1. (top) Coastline and bathymetry (ETOPO-5 [*National Oceanic and Atmospheric Administration*, 1988]) of Prydz Bay, East Antarctica. The diamond marks Davis. The Sørsdal Glacier (SG) resides on the southern edge of the Vestfold Hills (VH, shaded area). (bottom) Close-up of the bathymetry (m) around Davis with the fast-ice site (circle marked S1), the typical position of the midwinter fast-ice edge (dashed line) and the meteorological site at Davis (square).

on thermodynamic processes. At shallow ocean sites, where the bathymetry restricts horizontal advection of water masses, and effectively cuts off the region from relatively warm deeper waters [*Nunes Vaz and Lennon*, 1996] the effect of oceanic forcing is negligible compared to atmospheric forcing. In a previous study, *Heil et al.* [1996] showed that off Davis Station, Prydz Bay (68°35′S, 77°58′E, Figure 1) the influence of oceanic heat on the fast-ice evolution is small compared to that of atmospheric forcing. Here we focus on interactions between atmospheric conditions and fast ice, to gain an understanding of how sea ice responds to atmospheric variability. Although based on a single station, historic ice and meteorological records provide useful insight into long-term changes in fast-ice characteristics.

[5] First we present observational atmospheric and fast-ice data from Davis, and discuss how these fit with the large-scale changes observed in the region. This includes the analysis of annual data as well as seasonal data: Autumn (March to May), Winter (June to August), Spring (September to November), and Summer (December to February in the following year). The temporal variability in these parameters is analyzed and interactions between the two media are investigated.

2. Atmospheric Parameters

[6] Meteorological measurements at Davis have been taken by the Australian Bureau of Meteorology (BoM) from 1957 onward, except for October 1964 to March 1969 when the station was closed. Atmospheric pressure data were collected at a site 13.5 m above sea level until 01 February 1994, when it was relocated to a site 23.2 m above sea level. Data are presented as mean sea level pressures. All other surface meteorological data were collected at a site 15.0 m above sea level for the entire record. Instrumentation varied through the years but their accuracy adhered to the standards of the World Meteorological Organisation. Accuracies for individual measurements using the current sensors are ±0.04°C (temperature), ±0.1 hPa (pressure), ±0.5 m s^{-1} and ±1° (wind speed and direction), and ±0.001 m (solid precipitation). Meteorological measurements of surface air temperature, mean sea level pressure, wind speed and direction used here consists of 3-hourly surface observations summarizing 43 years (1957–1964 and 1969–2003). In addition, 27 years of daily data (1977–2003) for solid precipitation are included here. The length of these data series (more than 30 years of continuous data for all fields but solid precipitation) should provide meaningful trend analyses [*Comiso*, 2000; *Watkins and Simmonds*, 2000] although results need to be viewed in context of the time span analyzed. Our statistical evaluation is based on confidence intervals using the *t*-distribution [*Kreyszig*, 1988]. Trends that are significant at the 90%, 95% or 99% confidence levels are indicated in parenthesis.

[7] Annual data are only derived for years for which complete observations are available, and similar for seasonal data. Summer seasons span from the end of one calendar year into the next and are referenced to the calendar year in which they start.

2.1. Surface Temperature

[8] Using the longest continuous data record of 3-hourly observations at Davis, the 34-year trend (1970–2003) of the annual mean surface air temperature shows a slight warming of +0.003°C yr^{-1} (Figure 2, top). The analysis of this 34 year record for seasonal trends reveals warming during winter (significant at the 95% level) and spring (95), and cooling during summer (90) and autumn (Table 1). The extended (46 year) trend of annual air temperature (1958–2003, with a gap from 1964–1969) exhibits warming of +0.009°C yr^{-1} (90), while the trend of annual air temperature derived for a more recent, shorter time interval (1979–

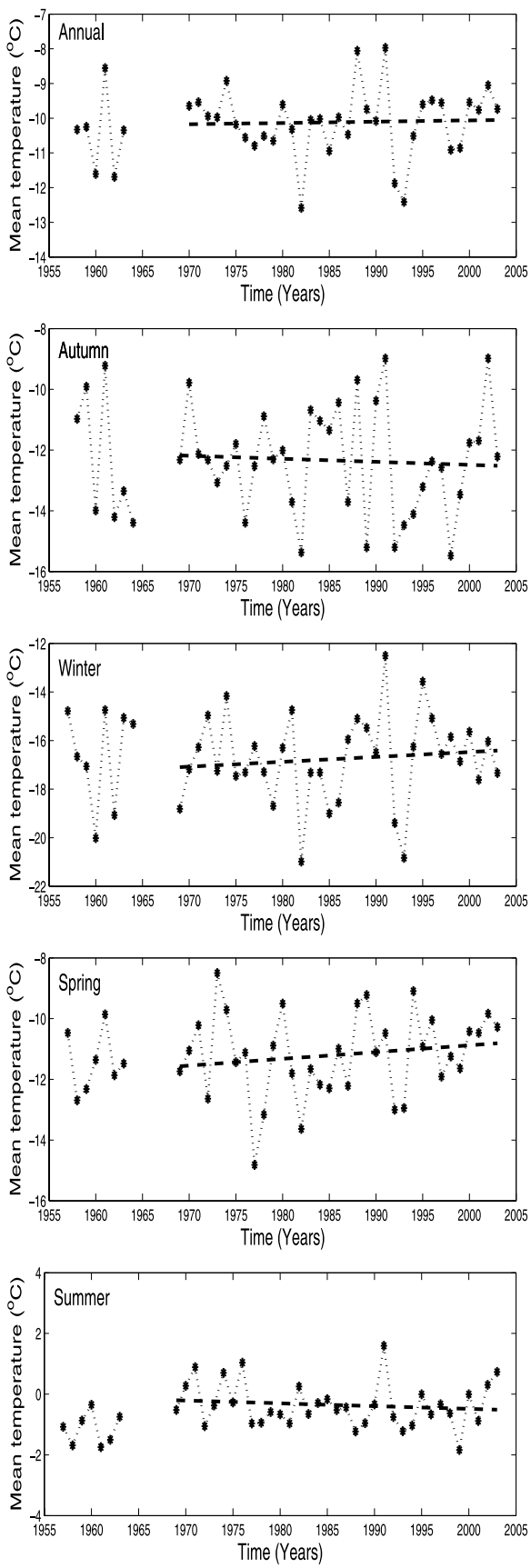

Figure 2. Mean annual (1958–1963 and 1970–2003) and autumn, winter, spring, and summer surface temperatures (for all available years) as observed at Davis. Trends (dashed line) are for the second interval only.

Table 1. Linear Trends in Mean Summer, Autumn, Winter, Spring, and Annual Surface Temperature at Davis for Two Time Series[a]

	Total Time Series	Continuous Time Series
Annual	(1958–1963 and 1970–2003) +0.009 (90)	(1970–2003) +0.003
Autumn	(1958–1964 and 1970–2003) −0.014	(1969–2003) −0.011
Winter	(1957–1964 and 1970–2003) +0.008	(1969–2003) +0.016 (95)
Spring	(1957–1963 and 1970–2003) +0.015 (90)	(1969–2003) +0.021 (95)
Summer	(1957–1963 and 1970–2003) +0.012	(1969–2003) −0.011 (90)

[a]Trends given in °C yr^{-1}. Significance levels at 90% and above based on the *t*-distribution are indicated.

1998; to coincide with satellite infrared observations by *Comiso* [2000]) shows an even larger warming trend (see below). This change in magnitude of the annual temperature trends calculated here using the same time series over different time intervals demonstrates that extreme care needs to be taken when using these to interpret changes.

[9] Comparison with the analysis of other investigators support the validity of our method of trend calculations. *Jacka et al.* [2004] used monthly data from 1959 to 2002 for the same location to derive a 44-year trend (1959–2002, with a gap from 1965–1968) of annual air temperature of +0.008°C yr^{-1}, which agrees well with our extended trend. Similarly our result derived from in situ observations is of the same magnitude as the temperature trend derived from remotely sensed data for 1979–1998 using Temperature Humidity Infrared Radiometer [*Comiso*, 2000]. The discrepancy between the 34-year and shorter (20 year) as well as longer (46 year) trends suggests that the annual temperatures are modulated by decadal or other long-term forcing, which causes interannual variations in the surface air temperature. A possible mechanism for the temporal variability in East Antarctic air temperature is discussed below.

[10] *Comiso* [2000] shows an annual warming trend of +0.012°C yr^{-1} for Davis, compared to +0.017°C yr^{-1} (95) derived from our data. The difference in the warming trend between the satellite-derived data and the surface observations used here is most likely due to the difference in the data's footprint (25 km by 25 km for the satellite-derived data versus single point station data). Because of their large footprint the satellite-derived data most certainly contain data influenced by the neighboring Sørsdal Glacier and the inland glacial ice, which are within the same pixel as the Davis data signal.

[11] The seasonal pattern of winter and spring warming, and summer and autumn cooling (identified in the continuous 35-year (1969–2003) record; Table 1) is also seen in NCEP/NCAR reanalysis data for that region. These synoptic, seasonally opposing trends might be a consequence of the Southern Hemisphere Annular Mode (SAM). The latter describes the leading mode of variability for extratropical cyclones at intermediate (weeks to months) timescales

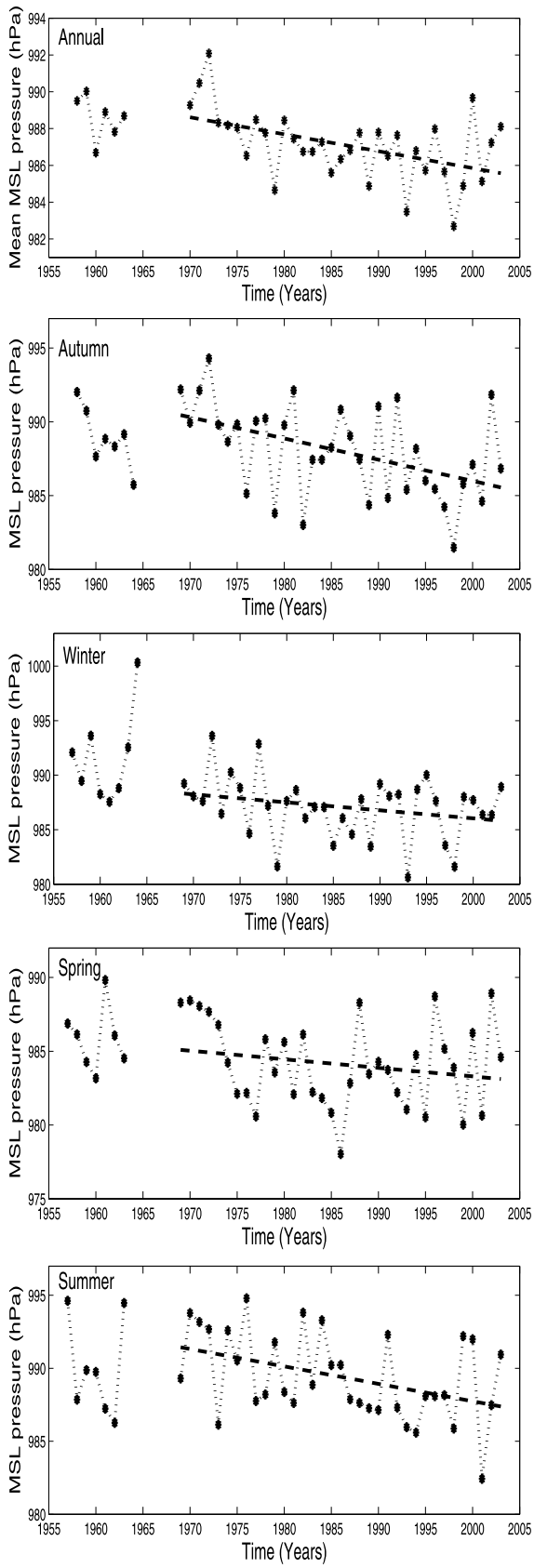

Figure 3. Mean annual (1958–1963 and 1970–2003) and autumn, winter, spring, and summer mean sea level pressure (for all available years) as observed at Davis. Trends (dashed line) are for the second interval only.

[*Thompson and Solomon*, 2002] and is driven by changes in the polar vortex. *Thompson and Solomon* [2002] suggest that the vertical coupling between troposphere and lower stratosphere is well established at times of perturbation in the polar vortex. Observations of the Antarctic atmosphere reveal a significant cooling in the lower stratosphere during spring only [*Trenberth and Olson*, 1989]. The cooling signal propagates downward into the troposphere, and arrives at the surface in summer or even later. Stratospheric cooling was first evidenced in the mid 1980s and has been associated with changes in stratospheric ozone levels. By the late 1990s a cooling of ≈10 K (as compared to 1985 data) had been identified for the months of October and November [*Randel and Wu*, 1999]. Observations show that in the 1990s the polar vortex survived throughout spring and into summer in both the troposphere and the stratosphere, longer than previously recorded [e.g., *Hurrell and van Loon*, 1994; *Waugh et al.*, 1999]. This evidence of stratospheric cooling and its downward propagation suggest that the summer cooling at Davis may be a signature of the SAM. The interpretation of low mean summer air temperatures observed prior to the data gap during the mid 1960s is not conclusive.

[12] The lowest surface air temperature measured at Davis was −40.7°C (27 April 1998). Maximum temperatures can reach above +10°C during summer. Some years (e.g., 1982, 1992 or 1993) were significantly cooler than the average. The interpretation of such events are difficult to infer from single-station data, however, similar changes are evident in temperature data averaged across the Antarctic continent: The cooling of 1982 and in the early 1990s are likely to be a consequence of the eruptions of the South American volcano El Chichon [*Angell and Korshover*, 1983] and Mount Pinatubo in the Philippines [*McCormick et al.*, 1995], respectively. Increased levels of stratospheric particulate in the wake of volcanic eruptions cause cooling at the Earth's surface.

[13] The temporal variability in the long-term signal of air temperature is examined to see whether temperature anomalies at Davis are part of a dynamic spatial pattern in the large-scale circulation. Assuming polar symmetry such a nonstationary pattern would repeat itself at a point location. A spectral analysis of the continuous 3-hourly record of air temperature from 1969 to 2003 was carried out after applying a low-pass filter [*Heil and Hibler*, 2002] with an upper limit of the passband of 0.375 cycles per day. The filter design is chosen to minimize errors outside the transition band.

[14] The resulting spectral analysis (not shown here) of the low-pass filtered surface air temperature from Davis exhibits peaks in spectral power at periods of 1 day and of about 1 and 5 years, where the annual signal dominates the power spectrum. The annual and daily modes are due to the changes in the incident solar radiation associated with the Earth's orbit and rotation. Oscillations at a frequency of about 5 years may be associated with changes in the large-scale atmospheric circulation. *Kienzle* [2000] suggests that such changes are derived from stationary oscillations.

2.2. Mean Sea Level Pressure

[15] A downward trend (−0.10 hPa yr^{-1}, (90)) in annual mean sea level pressure for the 34 year time series (1970–2003) is obtained at Davis (Figure 3). The 46-year trend

Table 2. Linear Trends in Mean Summer, Autumn, Winter, Spring, and Annual Mean Sea Level Pressure at Davis for Two Time Series[a]

	Total Time Series		Continuous Time Series	
Annual	(1958–1963 and 1970–2003)	−0.09	(1970–2003)	−0.10 (90)
Autumn	(1958–1964 and 1969–2003)	−0.10	(1969–2003)	−0.14 (90)
Winter	(1957–1964 and 1969–2003)	−0.13	(1969–2003)	−0.07
Spring	(1957–1963 and 1969–2003)	−0.05	(1969–2003)	−0.06
Summer	(1957–1963 and 1969–2003)	−0.07	(1969–2003)	−0.12 (90)

[a]Trends given in hPa yr^{-1}. Significance levels at 90% and above based on the *t*-distribution are indicated.

(for data from 1958–1963 and 1970–2003) is similar (−0.09 hPa yr^{-1}) despite the data gap although not significant at the 90% confidence level. Downward trends in mean sea level pressure have consistently been identified in seasonal data (Table 2).

[16] Figure 3 (top) also shows that there is less interannual variability in the annual mean sea level pressure during the 1980s than during the 1990s and early 2000s and the early part of the record (up until 1979). From station data alone the cause for this increased interannual variability from the 1990s to early 2000s cannot be identified.

[17] The annual maximum of mean sea level pressure occurs in summer. For the 1969–2003 time series the seasonal reduction in mean sea level pressure is most pronounced during summer and autumn (−0.12 hPa yr^{-1} and −0.14 hPa yr^{-1}, respectively).

[18] Frequency analysis (see section 2.1 for details) of the 35-year record (April 1969–December 2003) of the low-pass filtered mean sea level pressure from Davis exhibits the maximum signal at a period of 0.5 years, which expresses the semiannual surface pressure oscillation [*van Loon*, 1967]. Summer and winter maxima occur when the circumpolar low-pressure trough takes its most northerly position [*King and Turner*, 1997]. A secondary maximum in the power–spectral density occurs at a period of about 5 years, which is not as pronounced as the equivalent signal in the frequency spectrum of the surface air temperature.

[19] The interannual variability in the mean sea level pressure is studied by analyzing its frequency spectrum. For each year the power–spectral density is derived, and then integrated over frequency bands (0.100 to 0.125 cycles d^{-1}, 0.125 to 0.182 cycles d^{-1}, 0.182 to 0.333 cycles d^{-1}, and so on). Substantial interannual variability in the total amount of energy of the 3-hourly mean sea level pressure and its distribution over the various frequency bands (Figure 4) is found. Contributions associated with all frequency bands but the one for the 0.125 to 0.333 cycles d^{-1} band (the cyclone active band for extratropical cyclones [*Simmonds*, 1996]) show a decrease from 1969 to 2003.

[20] Winter mean sea level pressure for 35 years from 1969 onward (Figure 5) exhibits a net upward trend in the cyclone frequency over Davis, with some years containing higher cyclonic activity than preceding or succeeding years: 1998 stands out of this time series with about twice as much energy associated with cyclone-active frequencies than any other year. This coincides with anomalies in Antarctic sea-ice extent [*Hanna*, 2001] and fast-ice characteristics [*Massom et al.*, 2003]. We note that 1998 was the warmest year in the combined global land and sea-surface temperature record for 1856 to 2002 [*Jones and Moberg*, 2003].

[21] Increased cyclonic activity may be due to a shift of energy from other frequency bands, or from increased total energy within the system. Comparison of temporal evolution in the power–spectral density within the cyclone-active frequency band (3–5.5 days; $P^{fMSL}_{3.0-5.5}$) to those in the overall power–spectral density shows that increased cyclonic activity is associated with increased energy in the overall

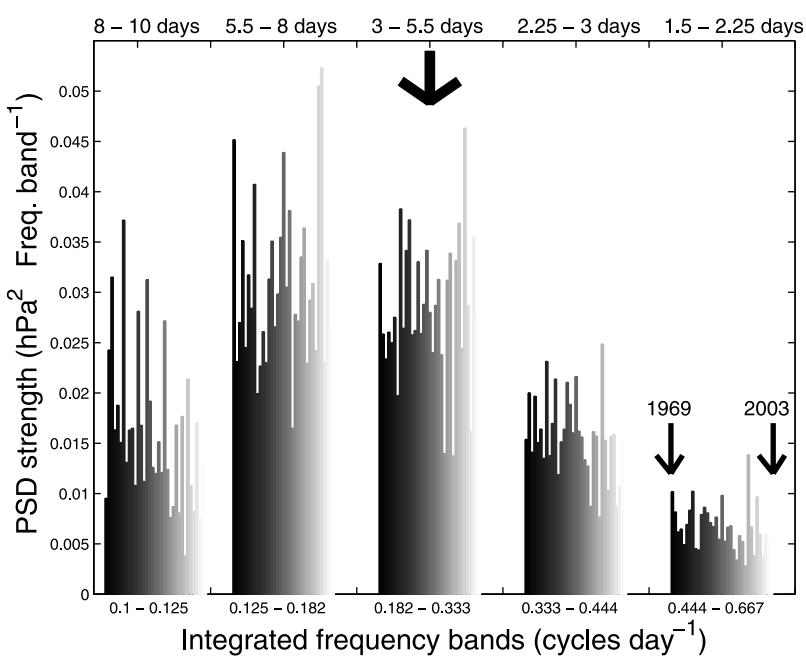

Figure 4. Integrated power–spectral density in multidaily frequency bands of winter mean sea level pressure at Davis. Different years (1969 to 2003) are coded in grey tones. The bold arrow indicates the frequency band capturing activity associated with low-pressure (cyclonic) systems.

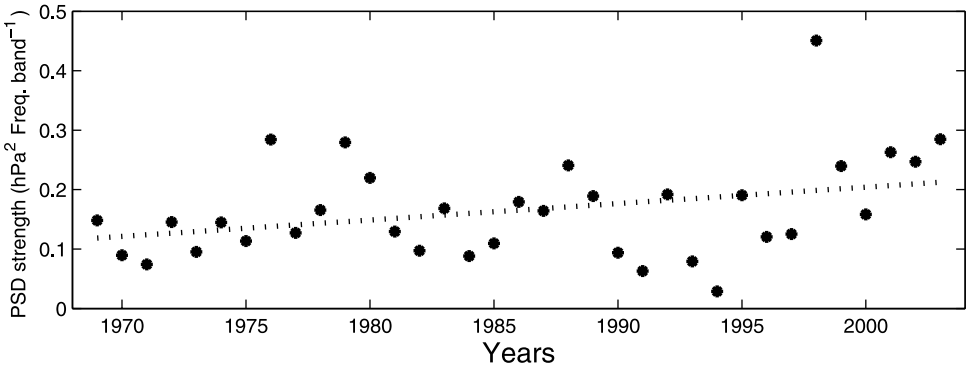

Figure 5. Integrated power–spectral density of winter mean sea level pressure for 1969 to 2003 in the frequency band from 3 to 5.5 days.

system. Despite a large bias in the surface pressure data of NCEP/NCAR 1 and artificial high-latitude trends [*Bromwich and Fogt*, 2004], these are used here to analyze the synoptic field. The NCEP/NCAR 1 data show a lowering in central cyclone pressure in the region of the Antarctic low-pressure trough. Analysis of trough intensity and distributions of mean and scatter of Davis mean sea level pressure give evidence that cyclonic activity increases during years of low mean sea level pressures ($R^2 = -0.57$, (90)). Together these indicate that the decrease in mean pressure is due to an increased number of cyclone systems and lower central pressures rather than to a general reduction in the horizontal pressure field.

[22] Vertical coupling throughout the troposphere [*Thompson et al.*, 2005] might provide a link for this kind of modification in cyclonicity with the changed seasonal pattern in growth and decay of the polar vortex (see previous section). The trend of winter warming in the lower atmosphere off Davis is a consequence of the delayed formation of the polar vortex, which is caused by the lengthened interval of southerly intrusions of relatively warm air associated with the winter cyclones. The results presented here agree with *Simmonds and Keay* [2000], who find an upward trend in cyclone deepening around the Antarctic continent in their analyses of NCEP surface data for 1948 to 1997.

2.3. Wind Speed and Direction

[23] While katabatic regimes are characteristic for many coastal locations around the Antarctic continent [*Connolley and Cattle*, 1994], Davis enjoys benign wind conditions, mostly due the presence of the low-lying Vestfold Hills that separate it from the Antarctic Plateau. The mean annual wind speed was 5.4 m s^{-1} for the interval from 1970 to 2003. The mean annual wind direction for this time interval is 69° clockwise positive relative to true north. The annual wind speed cycle exhibits a semi-annual signal (Figure 6), which corresponds to the one identified in the mean sea level pressure field. Together with the annual changes in mean wind direction, this confirms that Davis weather is largely driven by passing synoptic systems rather than by katabatic flow from the plateau.

[24] Figure 7 (top) shows an upward trend (+0.03 m s^{-1} yr^{-1}) in annual mean wind speed for the 34 year time series (1970–2003), obtained at Davis. The 46-year time series (1958–1963 and 1970–2003) exhibits a similar trend (+0.02 m s^{-1} yr^{-1}. The upward trend in mean wind speed is evident during all seasons (Figure 7). During summer both time series exhibit the strongest increase in wind speed: +0.04 m s^{-1} for 1969–2003 (90) and +0.03 m s^{-1} for 1957–2003 (see Table 3). These changes are likely to be associated with the strengthening of circumpolar eastward winds in the troposphere of the Southern Ocean region [*Hurrell and van Loon*, 1994; *Thompson and Solomon*, 2002].

[25] High wind speeds are infrequent compared to other locations around the East Antarctic coast. The highest 3-hourly wind speed (75 m s^{-1}) at Davis occurred on 30 July 1979, a year that exhibited several sustained high wind speed events. During the late 1990s the occurrence of high wind

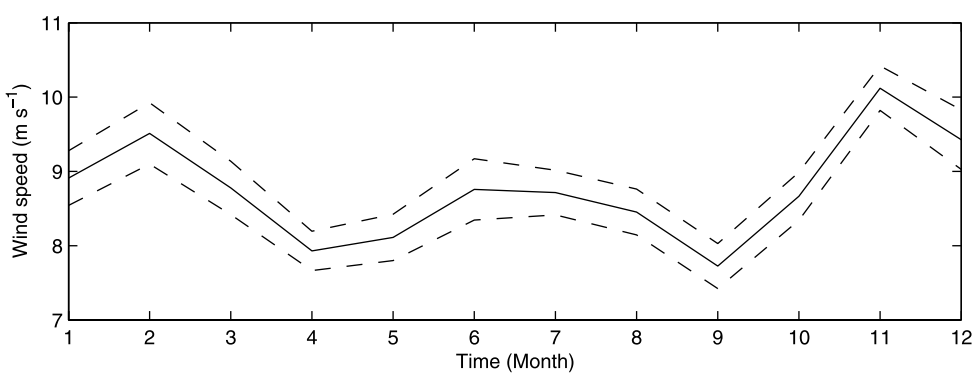

Figure 6. Monthly mean surface wind speed (solid line) at Davis (1957–1964 and 1969–2003). The 90% confidence interval is shown within dashed lines.

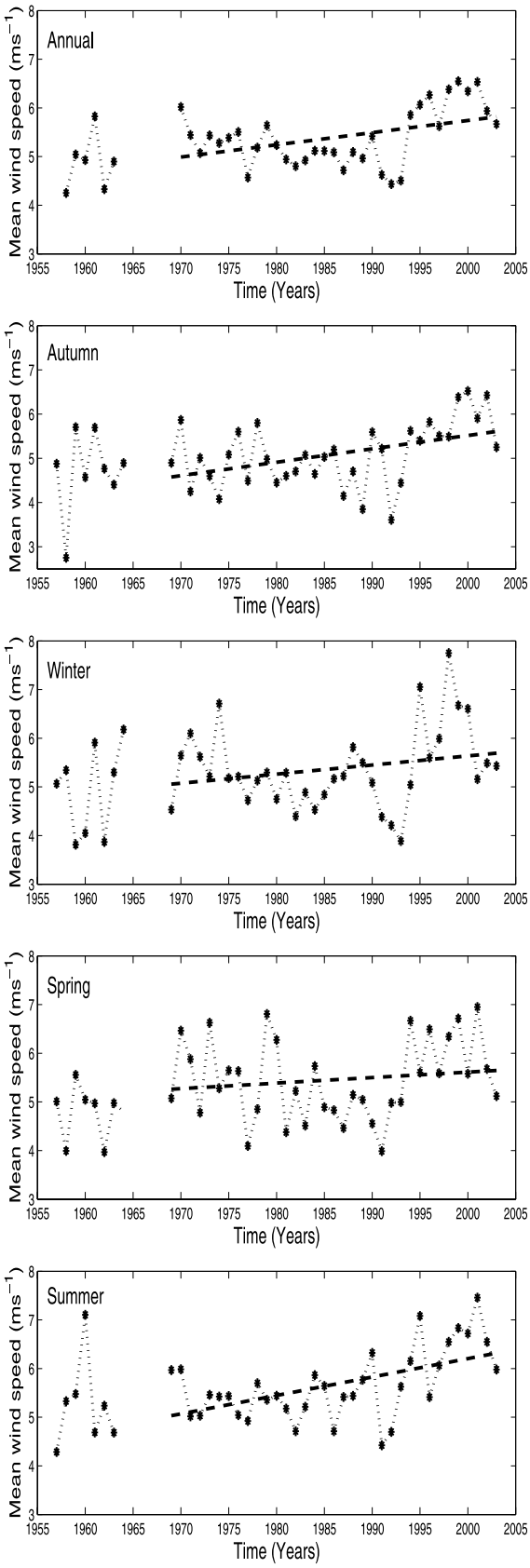

Figure 7. Mean annual and autumn, winter, spring, and summer 10-m wind speed as observed at Davis (1957–1964 and 1969–2003). Trends (dashed line) are for the interval from 1969 to 2003.

speed events was about twice the long-term average for the annual mean and the winter distribution (Figure 8). At Davis the annual series of high wind events (in excess of 20 m s^{-1} as derived from 3-hourly data) appears to be driven by multidecadal variability. Increased surface wind speed is strongly related to the increase in cyclonicity (R^2 = 0.83 for winter data, (90)).

[26] Surface winds at Davis are most frequently from directions between 40° and 85°. However, during summer there is a bias toward more northerly winds with surface winds frequently at directions from 10° to 60°. Mean annual wind direction tends farther toward South-Easterlies (+0.3° yr^{-1} for the 34-year trend from 1970 to 2003, (90)), with similar trends during autumn, winter and spring, and trend toward North-Easterlies for summer wind direction. The reversed summer trend in wind direction coincides with the timing of the largest tropospheric change in the Southern Hemisphere [*Thompson and Solomon*, 2002].

2.4. Precipitation

[27] Frontal cyclonic systems are the major source for precipitation near the Antarctic coast. Compared to other coastal sites, especially in Western Antarctica, precipitation at Davis is low: the long-term mean is 0.231 m water equivalent per year. The moisture transport is limited by a low saturation water-vapor pressure. This low saturation pressure is due to reduced moisture-loading capability of air at or below freezing temperature [e.g., *King and Turner*, 1997]. Analysis of the meteorological observations shows that at times when Davis is within the upstream region (on the easterly side) of a strong cyclonic system, heavy precipitation may precede or coincide with the arrival of a warm change. A further source of precipitation originates during intervals of open water (see section 3.4) from enhanced local evaporation. This mechanism is important during times when the ocean-atmosphere temperature gradient is large. Examples for such behavior in the local precipitation rates have been identified in the daily precipitation data from Davis.

[28] Solid precipitation is measured daily at Davis with a 0.203 m diameter snow gauge, which is housed inside a nipher shield. To avoid spurious precipitation data due to blowing snow, no measurements were taken when the wind speed is above 14 m s^{-1} [*Phillpot*, 1997]. High wind speeds prevented precipitation data being obtained about 5% of the time. An upward trend of +0.003 m yr^{-1} in total solid precipitation (Figure 9, top) has been found in the 26 year time series (1977–2002) of daily measurements. Increasing precipitation rates were observed for all seasons except summer. The spring trend is statistically significant at the

Table 3. Linear Trends in Mean Summer, Autumn, Winter, Spring, and Annual 10-m Wind Speed at Davis for Two Time Series[a]

	Total Time Series	Continuous Time Series
Annual	(1958–1963 and 1970–2003) +0.02	(1970–2003) +0.03
Autumn	(1958–1964 and 1969–2003) +0.02	(1969–2003) +0.03
Winter	(1957–1964 and 1969–2003) +0.02	(1969–2003) +0.02
Spring	(1957–1963 and 1969–2003) +0.03	(1969–2003) +0.02 (95)
Summer	(1957–1963 and 1969–2003) +0.03	(1969–2003) +0.04 (90)

[a]Trends given in m s^{-1} yr^{-1}. Significance levels at 90% and above based on the *t*-distribution are indicated.

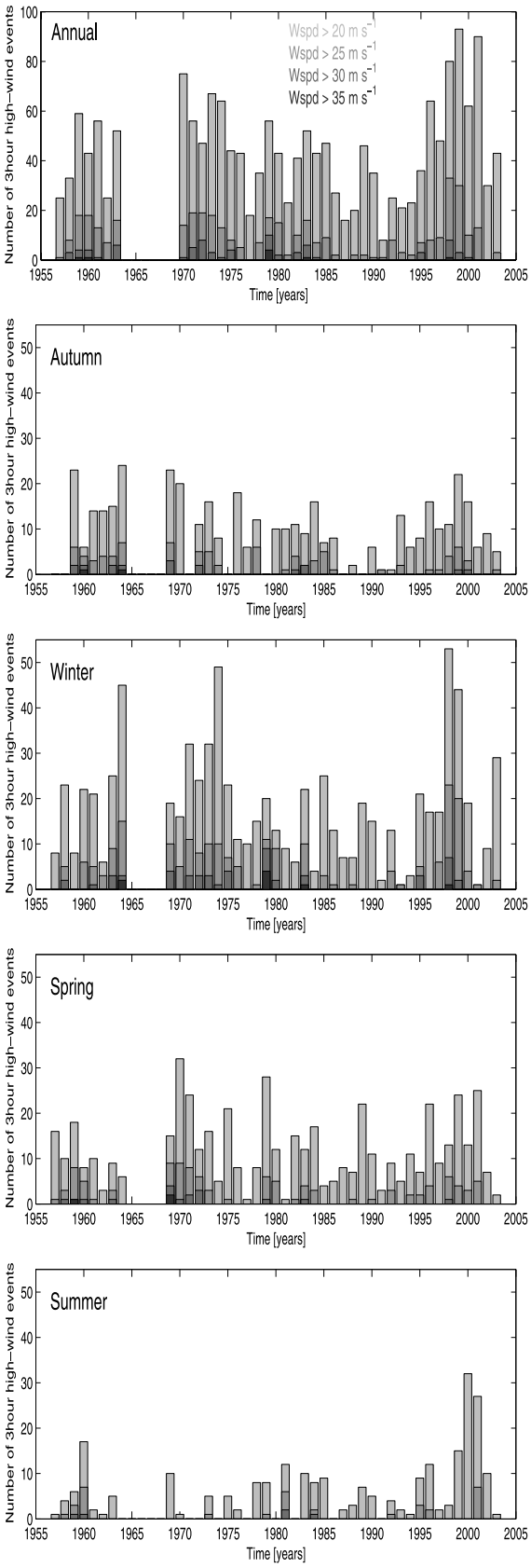

Figure 8. Mean annual and seasonal distribution of wind events with wind speeds (wpsd) above 20, 25, 30, and 35 m s^{-1} for 10-m winds as observed at Davis (1957–1964 and 1969–2003).

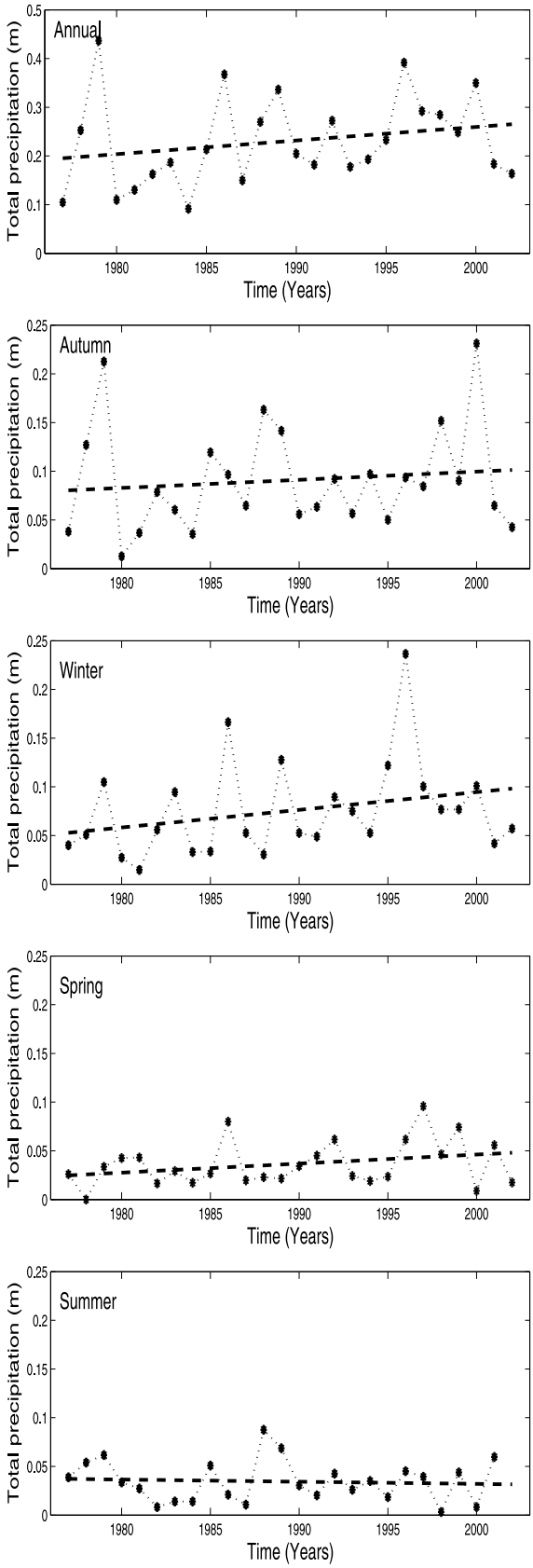

Figure 9. Net accumulated precipitation (liquid equivalent) for complete years and autumn, winter, spring, and summer from daily measurements at Davis (1977–2002) for wind speeds of 14 m s^{-1} or less.

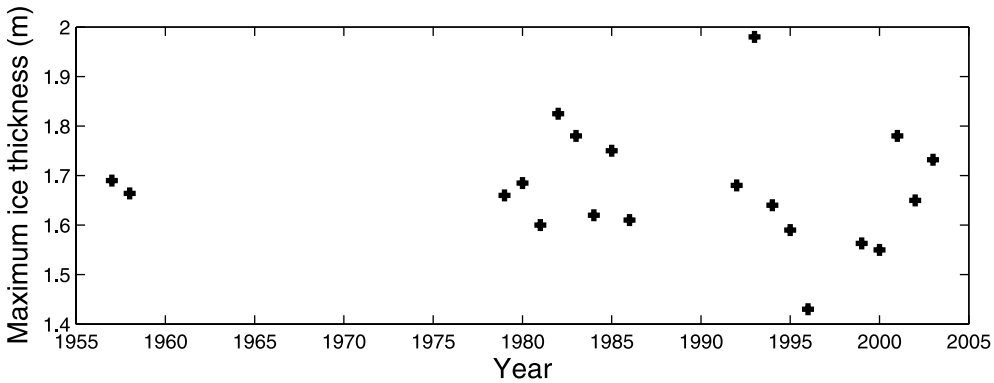

Figure 10. Annual maximum fast-ice thickness for 20 years from the Davis record.

90% confidence level, the annual and the winter trend are significant at the 95% confidence level. The summer measurements reveal a downward trend, but summer and autumn trends are statistically not significant.

[29] Although the observed increases in precipitation are in general agreement with changes predicted for warming scenarios [*Budd and Simmonds*, 1991], these changes appear too extensive to be solely due to the observed warming. Hence changes in the atmosphere dynamics were investigated as the origin for the precipitation changes. The annual mean of the total solid precipitation and, especially, the autumn total precipitation are highly anticorrelated with the mean sea level pressure. There is no correlation between precipitation and air temperature annually or seasonally. The increases of solid precipitation are likely to be influenced by the increased cyclone frequency, or by changes in the intensity of the cyclones.

3. Fast-Ice Parameters

[30] Time series of fast-ice parameters are useful to study interactions in the polar ocean-ice-atmosphere system and to identify their variability and trends. Here we analyze the interannual variability of fast-ice characteristics off Davis, where a seasonal fast-ice cover is encountered. With little or no perennial ice there is no direct history from previous years in the Davis fast-ice record although the fast ice is preconditioned by summer and autumn ocean and atmospheric conditions. The fast ice does not form keels and grounding is only observed within a couple of meters of the beach.

[31] In the immediate fast-ice area around Davis, also called Davis Harbour, maximum water depth is less than 30 m (Figure 1, bottom panel). Deeper water is found on the seaward side of the outer islands (150 m water depth at about 1 km distance), where transient fast ice is encountered. The latter does generally not survive into the melting season because of removal, for example, by passing icebergs. Owing to overall shallow ocean depths near Davis and a water depth of 12.4 m at the fast-ice sampling site (labeled S1 in Figure 1, bottom) it is presumed that after the onset of freezing the supply of oceanic heat within the surface mixed layer of Davis Harbour is negligible, and that the fast ice's energy budget is largely driven by atmospheric forcing [see also *Heil et al.*, 1996].

[32] In situ fast-ice measurements near Davis using drill holes are available intermittently since 1957. Sampling locations, frequency and accuracy of the measurements varied over the years. In this study data are used only from a single site (S1), and for years when measurement accuracy and sampling frequency were high (i.e., ±0.01 m or better; twice monthly or more frequent), and when human impact (e.g., through shipping movement) on the ice evolution can be excluded. Owing to these restrictions data from site S1 for years 1957, 1958, 1979 to 1986, 1992 to 1996, and 1999 to 2003, summarizing 20 seasons, are discussed in the following. Site S1 is about 1 km off the Davis coast, seaward of O'Gorman Rocks; the fast ice at this site is undeformed and free of any (tidal) cracks or snow drifts. Measurements of snow depth and freeboard were also obtained, as were visual observations of ice and ocean conditions, including the dates of ice formation and breakout. The primary focus is on the discussion of the annual maximum ice thickness, day of maximum ice thickness, breakout date, and duration of the fast-ice season.

3.1. Maximum Ice Thickness

[33] Near Davis the growth of the fast ice is determined by thermodynamic processes; ice advection or ice deformation do not contribute to the ice thickness. In the vicinity of Davis, the offshore fast ice is prone to repeated breakout early in the growth season, hence termed transient fast ice. However, the inshore fast ice is generally not affected by such breakout. The fast ice in the Davis Harbour, including Site S1, is not affected by the formation of snow ice, neither has any contribution of frazil ice been recorded at site S1. Frazil ice inclusion has been observed farther seaward, near the transient fast-ice edge. Fast-ice removal late in the season (during summer) generally occurs owing to advection, which typically is initiated by strong wind events, often in conjunction with extreme tides (king tides). By this time ablation has become the dominant process over ice growth and the fast-ice thickness has reduced from the annual maximum. Hence, at Davis, where the annual maximum thickness of the inshore fast ice generally occurs well before the final ice breakout, the annual maximum ice thickness reflects the integrated strength of the thermodynamic forcing, and hence may be used as a climate indicator.

[34] The long-term maximum occurred in 1993 (Figure 10) when the fast-ice thickness peaked at 1.98 m. The minimum in the annual maximum ice thickness occurred in 1996 (1.44 m). Using all 20 years of data the long-term mean

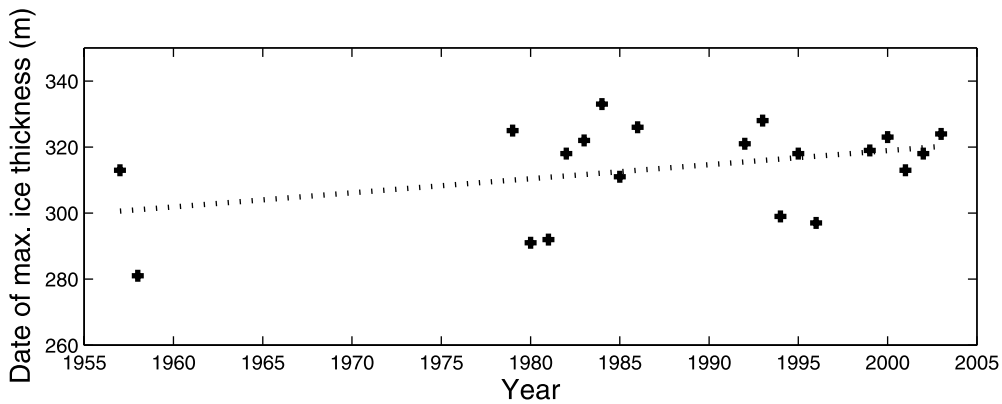

Figure 11. Date of annual maximum ice thickness at Davis. The long-term trend (dotted line) is positive.

annual maximum ice thickness (with standard deviation) was 1.67 (±0.11) m. This mean does not differ from the mean for data from 1979 onward: 1.67 m (±0.12 m). Our data suggest that the average annual maximum ice thickness has not changed significantly from the late 1950s to those of the most recent 24 years.

[35] The continuous record from 1979 to 1986 (8 years) shows no change (0.00 m yr^{-1}) in annual maximum ice thickness, similar to the noncontinuous record from 1992 to 2003 data (10 years of data; 0.00 m yr^{-1}). Over the shorter interval from 1992 to 1999 (6 years of data) a decrease of -0.04 m yr^{-1} (90) in maximum ice thickness has been identified. During the early 2000s the annual maximum ice thicknesses were close to the long-term average (2002) or above (2001 and 2003). Long fast-ice records for other Antarctic stations are rare. However, similar results have been found for Arctic fast ice [*Polyakov et al.*, 2003]: trends in annual maximum fast-ice thickness are small and are not statistically significant for four sites in the Kara, Laptev, Chukchi and East Siberian seas.

[36] The interannual variability of maximum ice thickness is characterized by large year to year changes rather than longer-term trends. There was a substantial increase in the interannual variability of the annual maximum ice thickness during the 1990s relative to the previous epoch ($\sigma_{1990s} = 2.2 \times \sigma_{1980s}$). Both the maximum (1993) and minimum (1996) of the measured record of annual maximum ice thickness occurred during the 1990s.

[37] The large interannual variability in maximum ice thickness suggests that the seasonal fast-ice thickness evolution depends on a number of external forcing parameters, including the snow cover overlying the fast ice. Interannual changes in those parameters may combine to result in large year to year changes in maximum ice thickness, and yet mostly balance each other over decadal or longer timescales. The effect of external parameters on the fast-ice thickness is presented in section 4.

3.2. Date of Annual Maximum Ice Thickness

[38] The date of occurrence of the annual maximum ice thickness of fast ice signals the equilibrium phase between thermodynamic ice growth and ice melt. It is an indicator of the seasonal evolution of the thermal energy balance between ocean, sea ice and atmosphere. At shallow water locations such as Davis, interannual variability in the date of annual maximum ice thickness mainly reflects interannual changes in the atmospheric surface forcing, which includes the radiative and turbulent heat fluxes.

[39] At Davis, annual maximum ice thickness usually occurs between mid October and late November (Figure 11). Ablation is generally from the top, although sub-layer erosion and subsequent collapse of the ice surface have been observed. The interannual variability of the date of annual maximum ice thickness is large. However, on the basis of all available years the trend toward delayed occurrence in the date of annual maximum ice thickness is +0.43 d yr^{-1}. Considering data only from 1979 onward, this trend reduces to +0.24 d yr^{-1} (90). Years with early occurrence of annual maximum ice thickness (such as 1980, 1994 or 1996) exhibited mild spring temperature. For example during 1994 and 1996 the mean spring temperature was more than 2°C warmer than during other years in the 1990s (Figure 2, fourth panel).

3.3. Final Fast-Ice Breakout

[40] The time of the final fast-ice breakout varies spatially, largely as a function of distance from the coast. Fast ice within narrow bays or fjords is protected from ocean swell and wind fetch, and remains in place longer than the fast ice within Davis Harbour. The latter is separated from the moving pack by a coastal island, Gardener Island. The fast ice within Davis Harbour, covering about 12 km^2, usually breaks out on the one day.

[41] Observations of the final breakout date show interannual variations, with a net delay in the final breakout date (Figure 12). The trend toward delayed breakout date is +0.43 d yr^{-1} for the complete data record, and +0.33 d yr^{-1} for 1979 and later years. During the late 1950s, final ice breakout occurred in mid December (1957: Day of Year (DoY) 355; 1958: DoY 351), while in the 1980s, 1990s and 2000s final breakout occurred between late December and February (mean: DoY 370). Interannual variability of the fast-ice breakout date is high all through the record. The Davis trends indicate that a delay in the fast-ice breakout date occurred during the 1980s. Intense cyclones or high cyclone frequency are often associated with the breakout of fast ice. Hence delayed breakout of the fast ice is most likely an indication of a weakening of the spring and summer cyclone density, which is expressed in the air pressure field (see section 2.2).

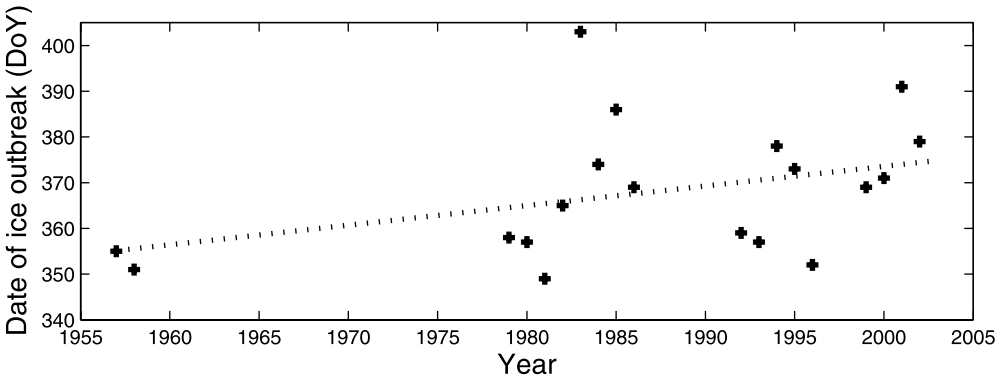

Figure 12. Date of the final fast-ice breakout at Davis. There is a long-term trend (dotted line) toward delayed breakout dates.

3.4. Annual Fast-Ice Duration

[42] The overall duration of fast-ice seasons is derived from data on the date of formation of stable fast ice and the date of the final ice breakout at the measurement site. Ice formation at Davis generally occurs simultaneously over a wide area (tens of km^2), including at the measurement site, whose distance from the coast of the Vestfold Hills is about 1.0 km.

[43] Occasional years of sustained fast-ice duration have been observed during the recent 2 decades, for example, 1983 or 1994 (Figure 13). The mean fast-ice duration has increased from 279 days in the late 1950s to 294 days in the 1980s, 300 days in the 1990s and 308 days in the 2000s. The long-term change (1950s to 2003) in duration is an increase of +0.67 d yr^{-1} (90). For years from 1979 to 2003, the increase in fast-ice duration is +0.77 d yr^{-1}.

[44] The change toward increased fast-ice duration exceeds the tendency toward delayed fast-ice breakout identified above. Interannual variability in the date of fast-ice breakout is larger, and dominates the interannual variability of fast-ice duration. Fast-ice formation at our measurement site occurred generally during early to mid March, except for 1994 to 1996 when ice formation took place in mid to late February. The date of fast-ice formation is largely determined by parameters such as summer warming of the ocean mixed layer, atmospheric cooling before and during ice formation, acting wind stress, and oceanic surface divergence. Initial (autumn) and final fast-ice breakout usually occurs owing to wind-driven ice removal associated with the passage of an intense low-pressure system and may be associated with cyclonic activity. Peak monthly tides are often found to coincide with fast-ice breakout.

4. Ice-Atmosphere Interactions

[45] Knowledge of the strength of ice-atmosphere interactions is required to extend our understanding of the processes taking place within the polar climate system. Here we study the physical interactions between components of the lower atmosphere and the fast ice using the Davis observations discussed above. Knowledge derived from studies such as this will assist our ability to numerically simulate and predict the interactive behavior of those components for various external forcing scenarios. While numerous and complex interactions take place between the lower atmosphere and the underlying fast ice, in the following only those linear interactions are presented which were found to be statistically significant at the 90% level or above.

4.1. Air Temperature Versus Annual Maximum Ice Thickness

[46] The thermodynamic ice growth at Davis depends largely on the temperature of the lower atmosphere. After the initial freezeup, the upper ocean is close to isothermal,

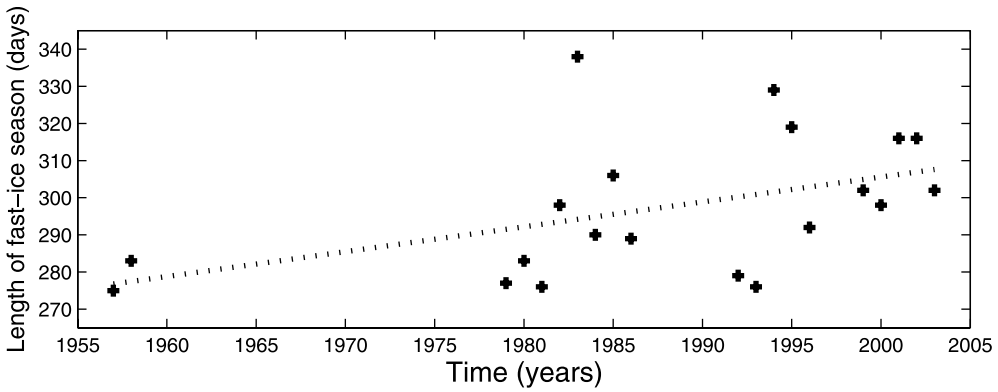

Figure 13. Fast-ice duration at Davis with a long-term trend (dotted line) toward prolonged annual ice cover.

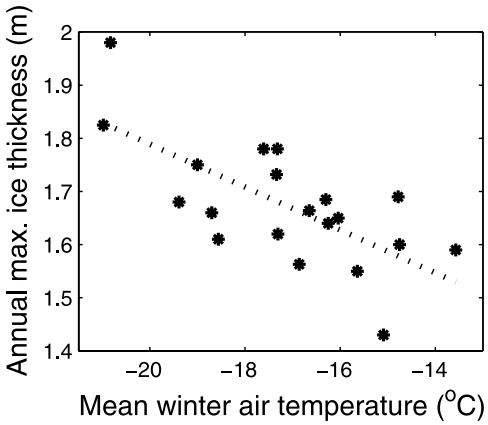

Figure 14. Scatterplot of mean winter surface air temperature and winter maximum ice thickness for 1957–1958, 1979–1986, 1992–1996, and 1999–2003.

and the vertical temperature gradient is largely determined by changes in the atmospheric temperature at the ice surface.

[47] On a year-to-year basis, the relationship between mean (annual and seasonal) surface temperature and maximum ice thickness becomes evident. For example, winter warming at Davis shows a high anticorrelation (Figure 14) with the annual maxima in fast-ice thickness ($R^2 = -0.69$) with a gradient of -0.041 m °C^{-1} (95). The correlation between the mean annual and spring air temperatures and the annual maximum ice thickness are similar ($R^2 = -0.63$, (95) and $R^2 = -0.64$, (95)).

4.2. Solid Precipitation Versus Annual Maximum Ice Thickness

[48] Solid precipitation and the resulting snow cover are examined as these also affect the ice-growth rate, and hence influence the annual maximum ice thickness. Although solid precipitation is generally redistributed throughout the season by winds, we find a good correlation ($R^2 = 0.58$, (90)) between cumulative precipitation and the annual maximum snow thickness measured at the fast-ice site. Any snow cover insulates the sea ice from the overlying atmosphere and changes the thermal conductivity of the ice-snow slab [e.g., *Maykut*, 1978]. The effect of snow on the thermodynamic growth of sea ice is bifold owing to the competing effects of its insulation: During the active growth season in autumn and winter the snow blanket slows the rate of ice growth, in spring and summer it protects the sea-ice surface from melt. At Davis the strongest signal between cumulative solid precipitation and annual maximum ice thickness occurs during winter, when they are anticorrelated at $R^2 = -0.47$ (Figure 15). The tendency toward lower ice thickness with increasing precipitation is -0.34 m m^{-1} (95%). Annual cumulative solid precipitation and annual maximum ice thickness behave similarly ($R^2 = -0.44$, (90)). The inverse relationship reflects the seasonal impact of a snow cover on the growing fast ice. Our data revealed no further relationship between snow cover and ice characteristics for other seasons.

4.3. Cyclonicity Versus Annual Maximum Ice Thickness

[49] The 35-year interval from 1969 to 2003 is largely characterized by downward trends in mean sea level pressure, which has been associated with an increased cyclonicity (section 2.2). Direct comparison of temporal changes in the mean annual or any seasonal mean sea level pressure relative to changes in the annual maximum ice thickness does not reveal any relationship between the parameters. However, a relationship is found when comparing changes in the annual maximum ice thickness to changes in the frequency-integrated cyclone-active power–spectral density of annual mean and winter mean sea level pressure. Correlation coefficients of $R^2 = -0.54$ and $R^2 = -0.48$ are obtained, respectively. The tendency in reduction of maximum ice thickness with increasing annual cyclone strength is -0.43 m (hPa^2 (cycles d^{-1})$^{-1}$)$^{-1}$ (90) (Figure 16).

[50] The relationship between the annual maximum ice thickness and winter $P^{fMSL}_{3.0-5.5}$ emphasizes the influence of passing cyclones on the growth conditions of the fast ice: The cyclones not only facilitate the breakup of fast ice along the outer edge, they also provide a poleward transport of moist and relatively warm air. This can induce local warming intervals. Surface measurements at Davis indicate that

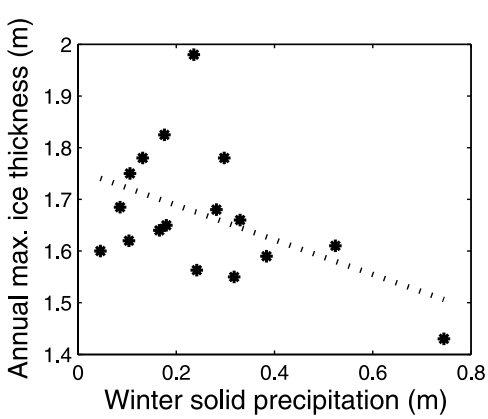

Figure 15. Scatterplot of cumulative solid precipitation during winter and annual maximum ice thickness for 1957–1958, 1979–1986, 1992–1996, and 1999–2003.

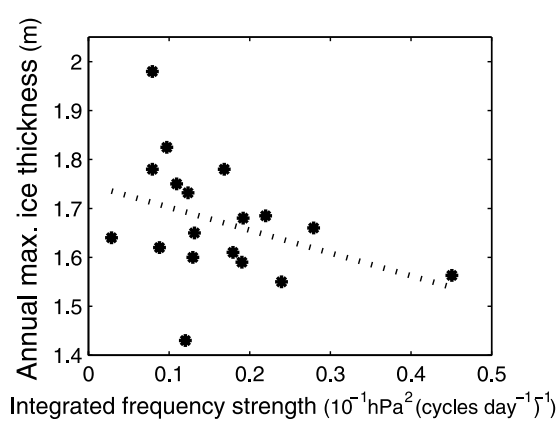

Figure 16. Scatterplot of annual maximum thickness of the fast ice as a function of the annual $P_{3.0-5.5}^{fMSL}$ can be derived at Davis for 1957, 1958, 1979–1986, 1992–1996, and 1999–2003. The tendency is also shown (dotted line).

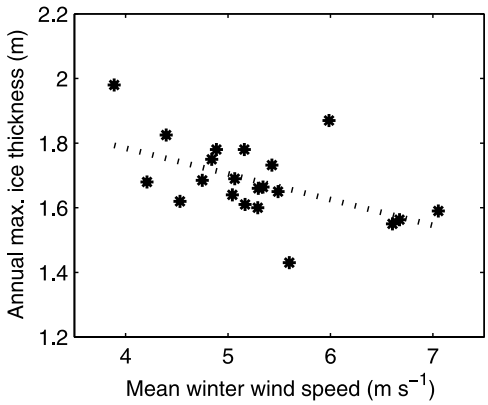

Figure 17. Annual maximum ice thickness versus mean winter wind speed at Davis for 1957, 1958, 1979–1986, 1992–1996, and 1999–2003. The tendency is also shown (dotted line).

southward advected air warms the area for intervals of up to 2 to 4 days before temperatures drop to cooler values. These warming events, especially during winter, may affect the fast ice by slowing the ice-growth rate.

4.4. Wind Speed Versus Annual Maximum Ice Thickness

[51] Increased cyclonicity and increased wind speed are closely correlated at Davis (section 2.3). Similar to cyclonic activity mean annual and winter wind speeds influence the annual maximum ice thickness (Figure 17). Correlation coefficients are $R^2 = -0.46$ (95) and $R^2 = -0.52$ (95), respectively. The tendency in reduction of maximum ice thickness with increasing mean winter wind speed is -0.08 m (m s^{-1})$^{-1}$. Wind speed is thought to influence the thermodynamic growth via the horizontal advection of warmer air, as suggested by the good correlation ($R^2 = +0.52$) between winter wind speed and air temperature.

4.5. Air Temperature Versus Date of Annual Maximum Ice Thickness

[52] Changes in the thermal forcing, especially in the seasonal redistribution toward warmer winters and cooler summers, affect the seasonal evolution of the fast ice, and are reflected in temporal changes in the ice-growth curve. Winter and spring warming delay the timing of the thermodynamic equilibrium state between ice growth and ice ablation. This lengthens the interval of active thermodynamic ice growth. Warming of the lower atmosphere during winter corresponds to a delay in occurrence of the annual maximum ice thickness by +0.46 days °C^{-1} (95). Winter warming correlates well with delays in annual maximum ice thickness ($R^2 = +0.47$). The correlation between spring warming and date of occurrence of the maximum ice thickness is also significant at the 90% level ($R^2 = +0.42$).

4.6. Cyclonicity Versus Breakout Date

[53] As discussed above, the breakout of the Davis fast ice is generally induced by strong winds associated with a passing low-pressure system. The relationship between the date of breakout of fast ice and changes in the mean sea level pressure is investigated for the annual as well as for the seasonal time series. Davis data show an inverse relationship between the breakout date and the summer cyclonicity (R = −0.59) with a tendency of −85.7 days (hPa^2 (cycles d^{-1})$^{-1}$)$^{-1}$ (90). Similar anticorrelations are found during winter ($R^2 = -0.52$, (90)) as well as the annual ($R^2 = -0.54$, (90); Figure 18) time series of mean sea level pressure. Increased cyclonic activity causes early occurrence of the ice breakout, while the fast ice persists longer into the summer when the integrated frequency strength in the cyclone-active band is low.

[54] It is furthermore worth noting the effect of high cyclonic activity during autumn and winter on the fast ice. When the new ice sheet forms and the ice is thin, the young fast-ice cover is easily removed during a storm. This has several effects on the physical properties of the fast ice as well as of the underlying ocean. Repeated removal of the fast ice increases the total ice production rate during a particular year, with impacts on the ocean as discussed before. Furthermore, the repeated removal of the fast ice delays the date of ice formation. A severe autumn or winter breakout of fast ice effectively reduces the annual maximum ice thickness obtained during that particular year, as the remainder of the growth season is not sufficient to make up for the loss of ice. Thin fast ice tends to breakout earlier than thicker fast ice. In consequence the coastal waters experience a prolonged open-water interval, which affects the growth of fast ice of the following year.

[55] While at Davis the measuring site is generally not affected by midwinter removal of the fast ice, during July 1998 major loss of near-coastal fast ice was observed. That winter exhibits the maximum cyclonic activity in the data record, and fast ice along the East Antarctic was short-lived [e.g., *Massom et al.*, 2003]. In recent years observations show that the midwinter fast-ice edge repeatedly approached to within a few kilometers of the coastline, while historical notes suggest that in the earlier years of observations the ice remained fast at about 8 to 10 km from the coast throughout the growth season. It appears that the annual fast-ice duration will reduce drastically in scenarios of increased cyclonicity.

5. Discussion

[56] In the previous section we presented those linear ice-atmosphere interactions, which are statistically significant at

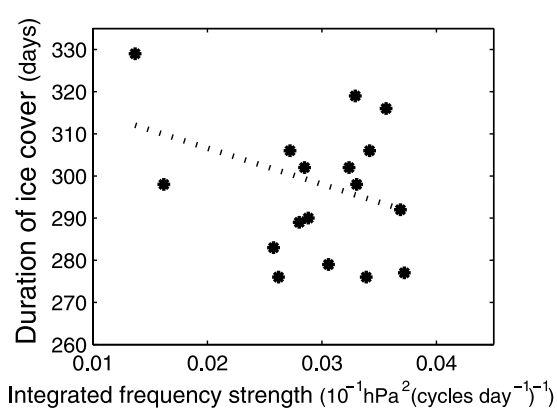

Figure 18. Duration of fast-ice versus annual $P_{3.0-5.5}^{fMSL}$ at Davis for 1957, 1958, 1979–1986, 1992–1996, and 1999–2003. The tendency is also shown (dotted line).

90% or above, based on the *t*-distribution [*Little and Rubin*, 1987]. A number of atmospheric parameters, including surface air temperature and wind speed, of those investigated were found to influence the annual maximum ice thickness. From our investigation we conclude that surface air temperature impacts on maximum ice thickness, while cyclonic activity and wind speed during autumn and winter appear have larger effect than accumulated precipitation on the growth of Davis fast ice.

[57] Nevertheless, all these atmospheric parameters relate to the fast-ice evolution through the thermal energy budget that describes the thermodynamic growth of sea ice [e.g., *Heil et al.*, 1996]. It is worth noting, that at locations where the oceanic heat flux is not negligible, oceanic contributions also affect the ice growth rate. On the basis of the analysis of our data record alone, it is not possible to rank the importance of the atmospheric parameters in influencing the fast-ice growth at Davis. A thermodynamic model of fast-ice growth, such as the one applied to investigate the interannual variability of the oceanic heat flux near Mawson [*Heil et al.*, 1996], would be well suited to address the interplay of the diverse ice-atmosphere and ice-ocean interactions, and to carry out a sensitivity study.

[58] Although determined by the thermodynamic energy balance and the same external forcing parameters as the annual maximum ice thickness, the timing of this maximum ice thickness shows significant correlation only with winter and spring surface air temperature. The date of fast-ice breakout seems to depend largely on the cyclonic activity. No significant correlation has been found between the date of ice formation and any of the atmospheric parameters.

[59] One may speculate about the fate of fast ice (or sea ice in general) in a warmer and more perturbed polar atmosphere with a stronger westerly flow, as one may expect on the basis of the findings of *Thompson and Solomon* [2002]. Our results, although derived from a point location, suggest that the response in seasonal fast-ice evolution to atmospheric changes (such as winter warming) will be multilayered. The changes induced in the fast ice are likely to enter into further feedback processes with components of the polar climate system. For example, the surface warming and the changed atmospheric circulation lead to a prolonged ice-growth season without causing net change in fast-ice thickness. From this we deduce that the growth rate of the fast ice has reduced, which in turn induces change to other processes: A change in the rate of brine rejection into the ocean mixed layer is expected as the ice growth slows. This in turn might affect the onset of deep-water formation. Numerical simulations with a thermodynamic fast-ice model or a coupled ice-ocean model would be suitable to investigate such scenarios of changes in high-latitude conditions.

6. Conclusion

[60] The analysis of meteorological and fast-ice measurements from Davis revealed long-term trends as well as interannual change in atmosphere and sea-ice variables. For the interval from 1969 to 2003 atmospheric winter and spring warming are more pronounced than summer and autumn cooling. For the same interval the annual mean surface air temperature shows only a small upward trend. The mean sea level pressure undergoes a net downward change in its annual mean and during all seasons. Pressure trends for the autumn and summer interval for 1969–2003 are largest. We showed that reduced air pressure is associated with increased occurrence and amplitude of cyclonic systems rather than with a general reduction in the two-dimensional mean sea level pressure field. Although Davis surface pressure exhibits large interannual variability in the cyclone active band, a net increase in cyclonic activity has been identified. Statistically significant trends in surface wind speed are upward. The increase in wind speed appears to be stronger in summer than during other seasons.

[61] In situ data available intermittently from the 1950s to the early 2000s used in this study revealed that interannual variability of fast-ice characteristics has increased during the 1990s reflecting changes in atmospheric conditions. There has been no net change in annual maximum ice thickness over the long term. However, annual maximum ice thickness was reduced in the second half of the 1990s. A shift of $+0.43$ d yr^{-1} toward later dates in the occurrence of annual maximum ice thickness took place between the late 1950s and the mid 1980s. This parameter remained stable during the 1990s. High interannual variability governed the fast-ice breakout date and the annual duration of the fast-ice cover with a net delay of $+0.43$ d yr^{-1} in the date of fast-ice breakout.

[62] At Davis the response of the fast ice to changes in the local atmospheric conditions was identified. Our data indicate that changes in surface air temperature and cyclonic activity are reflected in a suite of changes across several fast-ice parameters. Change in surface air temperature was the quantity that most influenced the annual maximum ice thickness as well as its date of occurrence. Lower, and negative, correlations were found between the annual maximum ice thickness and precipitation, cyclonic activity, and wind speed. This study highlights the dependency of fast-ice properties on local meteorological conditions and hence provides an indication of the changes that may be expected in fast-ice characteristics in consequence of changes in polar atmospheric conditions.

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